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Anatomy
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1349

Motion and forces

French mathematician Nicolas d'Oresme worked out a new way of drawing graphs to represent the motion of moving objects. The graphs helped to explain the relationship between the speed, time, and distance traveled.

1368

Guild of Surgeons

The foundation of a Guild of Surgeons in England, in 1368, was the first attempt to provide rules and regulations for the profession. Before this, anyone—commonly barbers—had been able to practice surgery.



An array of surgical instruments illustrated in the manuscript *De Chirurgia* (On Surgery) by the great Arab physician Albucasis.



In 1357, French philosopher Jean Buridan developed the theory of impetus, the force that makes an object move.

1380

Rocket warfare reaches Europe

The first recorded use in Europe of rockets in warfare came at the battle of Chioggia, a naval conflict between the Italian cities of Venice and Genoa. Rockets are difficult to make and their military use showed how a greater understanding of gunpowder weaponry was developing.



1345

1385

1364

Astronomical clock

Italian clockmaker Giovanni de Dondi completed his astrarium—a complex clock with dials showing the movements of the Sun, Moon, and planets. It had more than 100 gear wheels. As well as allowing astronomers to calculate the position of heavenly bodies, it provided a calendar of Church holy days.

The astrarium had seven dials and its central weight swung around 30 times a minute.

“One could by this believe that the earth and not the heavens is so moved, and there is no evidence to the contrary.”

Nicolas d'Oresme on the rotation of Earth, *Livre du ciel et du monde*, 1377

1377

A rotating Earth

In his *Livre du ciel et du monde* (Book of Heaven and Earth), Nicolas d'Oresme disproved all the popular ideas that Earth was stationary at the center of the solar system. He also suggested that Earth rotated on its axis. However, he could not go as far as believing that Earth moved around the Sun.



Nicolas d'Oresme seated by an armillary sphere, a model of the solar system

